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import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Function for High-Boost Filtering
def high_boost_filter(image, boost_factor):

    kernel_size = 3

    # Averaging kernel
    kernel = np.ones((kernel_size, kernel_size),
                     dtype=np.float32) / (kernel_size ** 2)

    # Blur the image
    blur_image = cv2.filter2D(image, -1, kernel)

    # High-boost mask
    sharpened = image + (image - blur_image) * boost_factor

    # Convert values to valid range
    sharpened = np.clip(sharpened, 0, 255).astype(np.uint8)

    return sharpened

# Read image
image = cv2.imread("lab_image.jpg")

# Check image
if image is None:
    print("Image not found!")
else:
    # Apply High-Boost Filtering
    sharpened_image = high_boost_filter(image, 1.5)

    # Save image
    cv2.imwrite("sharpened_image.jpg", sharpened_image)

    # Display images
    print("Original Image")
    cv2_imshow(image)

    print("High-Boost Sharpened Image")
    cv2_imshow(sharpened_image)
```

Original Image



High-Boost Sharpened Image



